Part 1: Data Exploration and Visualization (20 marks)

In [72]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import warnings
warnings.filterwarnings('ignore')
```

In [73]:
```
path = '/Users/elzaniyazbekova/Desktop/ML/lifesat.csv'
df = pd.read_csv(path)
```

Q2 (5 marks) Print basic info and summary statistics .

In [74]:
```
df.describe
```

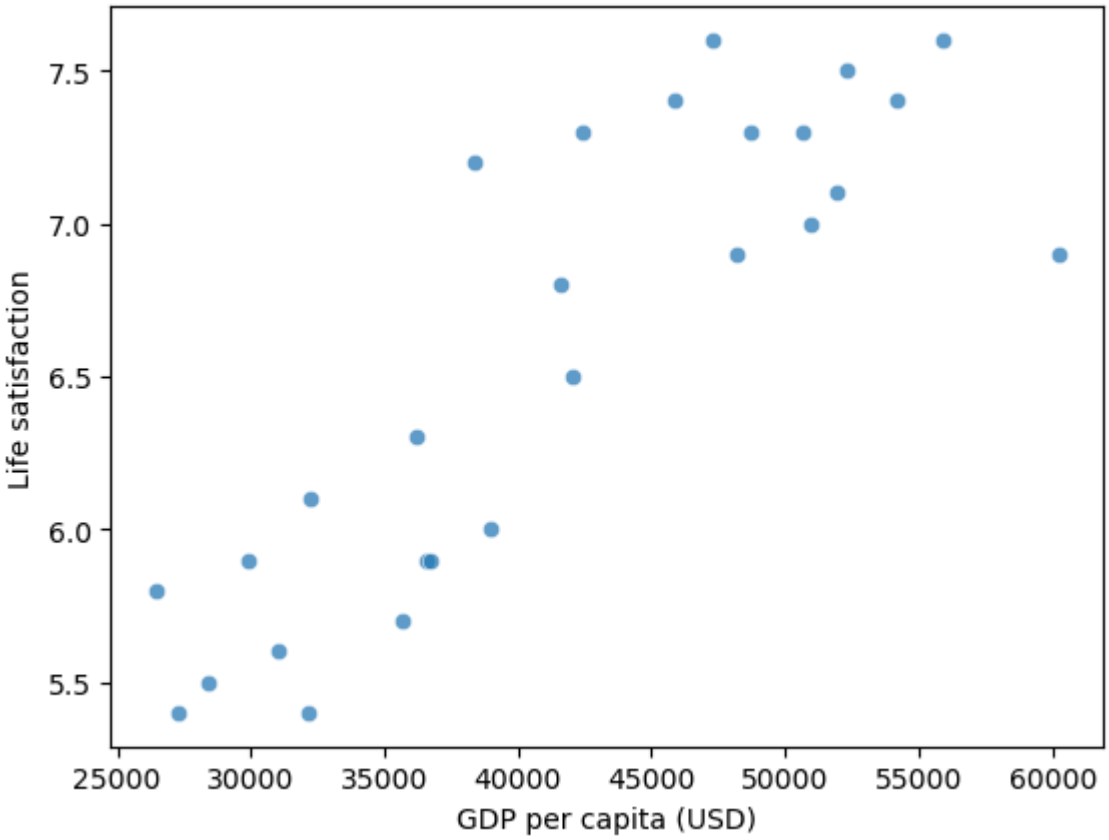
Out[74]:
```
<bound method NDFrame.describe of            Country  GDP per capita (US
D)  Life satisfaction
0           Russia          26456.387938                5.8
1           Greece          27287.083401                5.4
2           Turkey          28384.987785                5.5
3           Latvia          29932.493910                5.9
4          Hungary          31007.768407                5.6
5         Portugal          32181.154537                5.4
6           Poland          32238.157259                6.1
7          Estonia          35638.421351                5.7
8            Spain          36215.447591                6.3
9         Slovenia          36547.738956                5.9
10       Lithuania          36732.034744                5.9
11          Israel          38341.307570                7.2
12           Italy          38992.148381                6.0
13  United Kingdom          41627.129269                6.8
14          France          42025.617373                6.5
15     New Zealand          42404.393738                7.3
16          Canada          45856.625626                7.4
17         Finland          47260.800458                7.6
18         Belgium          48210.033111                6.9
19       Australia          48697.837028                7.3
20          Sweden          50683.323510                7.3
21         Germany          50922.358023                7.0
22         Austria          51935.603862                7.1
23         Iceland          52279.728851                7.5
24     Netherlands          54209.563836                7.4
25         Denmark          55938.212809                7.6
26   United States          60235.728492                6.9>
```

In [90]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27 entries, 0 to 26
Data columns (total 3 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Country               27 non-null     object
 1   GDP per capita (USD)  27 non-null     float64
 2   Life satisfaction     27 non-null     float64
dtypes: float64(2), object(1)
memory usage: 780.0+ bytes
```

Q3 (10 marks) Display a scatter plot for GDP per capita vs Life Satisfaction . Add labels, title, and discuss the observed relationship

In [76]:
```
sns.scatterplot(x='GDP per capita (USD)', y='Life satisfaction', alpha=0.
plt.show()
```



Discuss the observed relationship

Observed relationship takes on deviation, and is skewed positivly As wealth levels increase, the spread of the data indicates the level of other factors.

Part 2: Linear Regression Model (30 marks)

Q4 (5 marks) Extract input (X) and target (y). Print their shapes.

In [88]:
```
x = df[["GDP per capita (USD)"]]
y = df["Life satisfaction"]
print(x.shape, y.shape)
```

```
(27, 1) (27,)
```

Q5 (10 marks) Train a Linear Regression model & Display coefficient and intercept.

In [78]:
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(x, y)

print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
```

```
Coefficient: 6.778899694341226e-05
Intercept: 3.7490494273769075
```

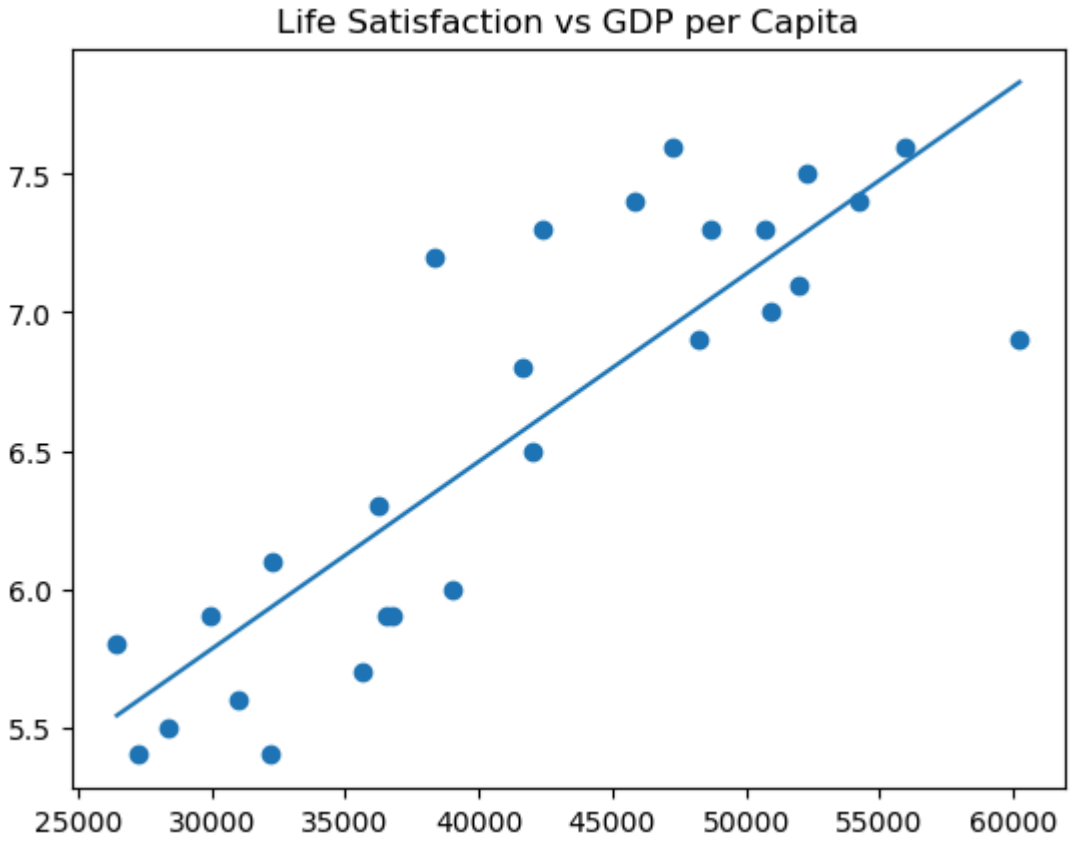
Q6 (10 marks) Plot the predicted regression line from the model along with a scatter plot of the data.

In [79]:
```
import matplotlib.pyplot as plt
import numpy as np


plt.scatter(x, y)
plt.plot(x_line, y_line)


plt.title("Life Satisfaction vs GDP per Capita")

plt.show()
```



Q7 (5 marks) Predict Life Satisfaction for GDP = 37,655.2 USD. Comment on result.

In [96]:
```
gdp_value = [[37655.2]]
predicted_satisfaction = model.predict(gdp_value)[0]

print(f"Prediction for GDP ${gdp_value[0][0]:,.2f}: {predicted_satisfacti
```

```
Prediction for GDP $37,655.20: 6.3016576650804845
```

Part 3: K-Nearest Neighbors Regression (25 marks)

Q8 (5 marks) Train a KNeighborsRegressor (n_neighbors=3).

In [92]:
```
knn_model = KNeighborsRegressor(n_neighbors=3)
knn_model.fit(x, y)
print(knn_model)
```

```
KNeighborsRegressor(n_neighbors=3)
```

Q9 (10 marks)Predict Life Satisfaction for GDP = 37,655.2 USD and compare with Linear Regression.

In [94]:
```
X_new = [[37655.2]]
knn_prediction = knn_model.predict(X_new)[0]
lip = model.predict(X_new)[0]
```

```
print(f"Linear Regression Prediction: {lip}")
print(f"KNN (k=3) Prediction:{knn_prediction}")
```

```
Linear Regression Prediction: 6.3016576650804845
KNN (k=3) Prediction:6.333333333333333
```

In [99]:
```
diff = knn_prediction - lip
print(f"Difference: {diff}")
print(f"KNN (k=3) Prediction:{knn_prediction}")
print(f"Linear Regression Prediction: {lip}")
```

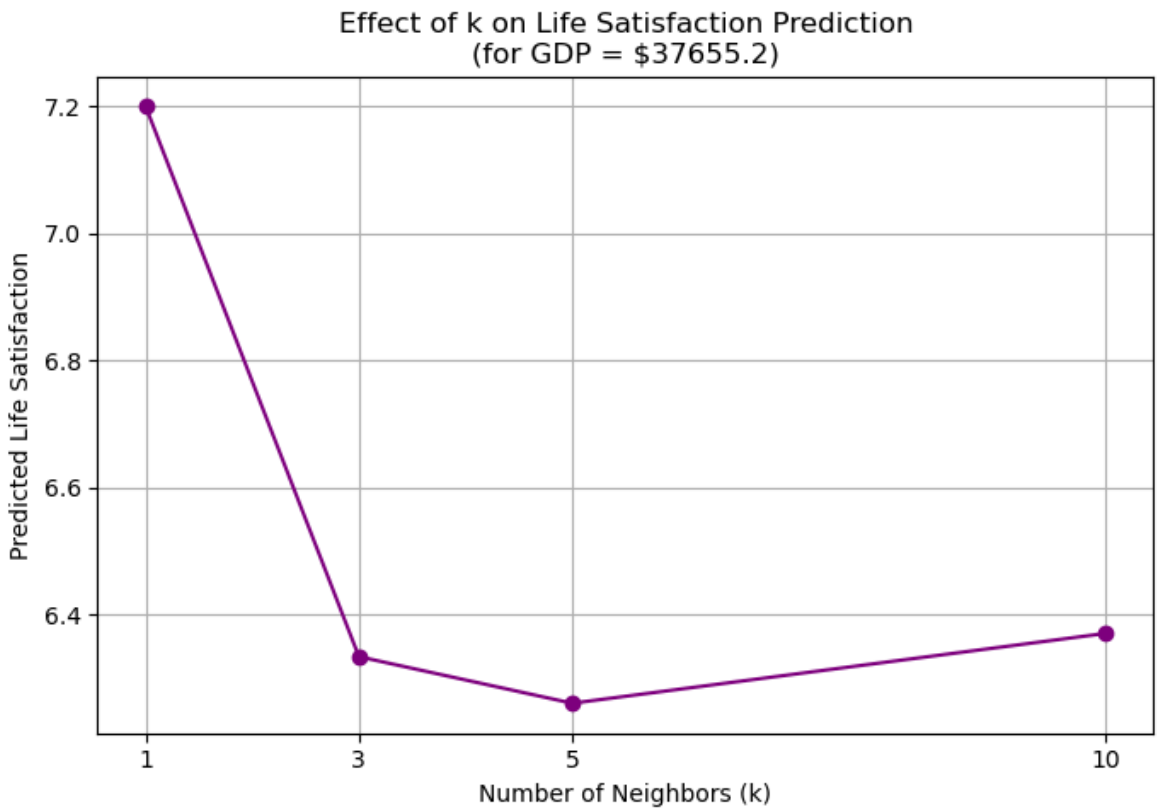
```
Difference: 0.03167566825284851
KNN (k=3) Prediction:6.333333333333333
Linear Regression Prediction: 6.3016576650804845
```

Q10 (10 marks) Use n_neighbors 1, 3, 5, and 10 and print the predicted values of life satisfaction in Q10. Plot the results using a line plot

In [104…
```
k_values = [1, 3, 5, 10]
predictions = []

for k in k_values:
    knn_k = KNeighborsRegressor(n_neighbors=k)
    knn_k.fit(x, y)
    pred = knn_k.predict(X_new)[0]
    predictions.append(pred)
    print(f"k={k}: {pred}")
plt.figure(figsize=(8, 5))
plt.plot(k_values, predictions, marker='o', linestyle='-'
, color='purple')
plt.title(f'Effect of k on Life Satisfaction Prediction\n(for GDP = ${X_n
plt.xlabel('Number of Neighbors (k)')
plt.ylabel('Predicted Life Satisfaction')
plt.grid(True)
plt.xticks(k_values)
plt.show()
```

```
k=1: 7.2
k=3: 6.333333333333333
k=5: 6.26
k=10: 6.369999999999999
```


Effect of k on Life Satisfaction Prediction
(for GDP = $37655.2)
Predicted Life Satisfaction
7.2
7.0
6.8
6.6
6.4
1
3
5
10
Number of Neighbors (k)